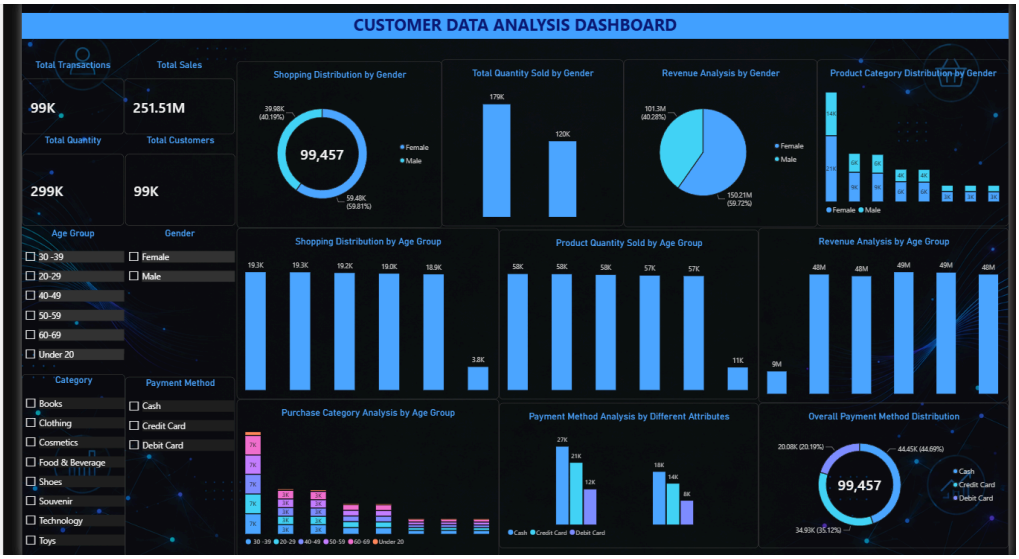
CUSTOMER DATA ANALYTICS

Analysis Report | Prepared from the supplied Power BI dashboard and 11 task questions



1. Objective and Dashboard Overview

The objective is to analyse customer shopping activity, product quantity, revenue, product categories, age groups, and payment methods using the supplied dashboard. The dashboard displays four headline KPIs: approximately **99K transactions**, **251.51M sales**, **299K quantity sold**, and **99K customers**. The screenshot does not show the currency or reporting period.

2. Answers to the Task Questions

Q1. How is the shopping distribution according to gender?

Female customers represent approximately **59.81% (59.48K)** of the shopping distribution, while male customers represent **40.19% (39.98K)**. Female customers therefore account for the larger share of shopping activity shown.

Q2. Which gender did we sell more products to?

The Total Quantity Sold by Gender chart shows a higher quantity for female customers (about **179K**) than male customers (about **120K**). Female customers are the larger segment by quantity sold.

Q3. Which gender generated more revenue?

Female customers generated approximately **150.21M (59.72%)** compared with **101.30M (40.28%)** for male customers. The dashboard therefore indicates that females contribute more revenue.

Q4. How are purchase categories distributed by gender?

The category-by-gender chart compares Clothing, Cosmetics, Food & Beverage, Toys, Shoes, Souvenir, Technology, and Books. Clothing has the largest visible total; Cosmetics and Food & Beverage are also prominent. The screenshot does not make every gender-by-category value readable, so exact category splits should be checked in Power BI.

Q5. How is shopping distributed by age?

The 20–29, 30–39, 40–49, 50–59, and 60–69 groups each appear to have roughly 19K shopping activity. The Under 20 group is much smaller, at approximately 3–4K. Most displayed adult age groups are broadly similar.

Q6. Which age category did we sell more products to?

The dashboard shows broadly similar shopping volumes for the adult age groups and a much lower value for Under 20. From the screenshot alone, it is not reliable to name one adult age group as the definite leader. Check the exact quantity values in the underlying visual or data table before stating a winner.

Q7. Which age category generated more revenue?

Revenue is much lower for Under 20, while the displayed adult age groups are at broadly similar levels. The screenshot does not support a confident ranking of the adult groups; verify the exact data labels or underlying values to identify the highest-revenue age group.

Q8. How are purchase categories distributed by age group?

The age-group category chart compares the same product categories across age groups. Clothing has the largest visible overall category total, with Cosmetics and Food & Beverage also contributing noticeably. Use the slicers or underlying data to identify the leading category within each age group; exact segment values are not legible in the screenshot.

Q9. Does payment method have a relationship with other columns?

The dashboard includes a payment-method-by-gender chart, so it allows a visual comparison of Cash, Credit Card, and Debit Card across Female and Male customers. The displayed differences suggest payment usage may vary by gender, but a chart alone does not prove a statistically significant relationship. To investigate further, compare payment method by age group, category, sales value, and transaction count, and use a cross-tabulation or suitable statistical test.

Q10. How is the payment method distributed?

Payment method	Approx. count	Share
Cash	44.45K	44.69%
Credit Card	34.93K	35.12%
Debit Card	20.08K	20.19%

Cash is the largest payment method (44.69%), followed by Credit Card (35.12%) and Debit Card (20.19%). Cash and Credit Card together account for about 79.81% of the displayed payment distribution.

Q11. Visualize the data and derive insights and suggestions for the company

The Power BI dashboard visualizes KPIs and comparisons using cards, doughnut/pie charts, bar charts, and a treemap. Its slicers allow users to filter by age group, gender, product category, and payment method. The key insights and recommended actions are summarized below.

3. Key Insights

• Female customers lead in shopping distribution, quantity sold, and revenue in the displayed dashboard.

• Most adult age groups have similar shopping volumes; Under 20 is the smallest group in the age visuals.

• Clothing is the most prominent category in the visible category charts; Cosmetics and Food & Beverage are also notable.

• Cash is the most common payment method, with credit cards second and debit cards third.

• The dashboard supports segment comparisons, but exact rankings should be verified where labels are too small to read.

4. Recommendations to the Company

• Run controlled marketing tests for high-contribution customer segments and measure incremental revenue rather than assuming that current shares will persist.

• Compare average order value, repeat-purchase rate, and product quantity by age group before choosing an age-based campaign.

• Monitor Clothing inventory and sales, but compare gross margin and stock turnover as well as quantity before increasing stock.

• Keep cash checkout convenient and ensure card payments are reliable; evaluate digital-payment promotions using cost and conversion measures.

• Add a date/month trend view and KPI definitions so the business can distinguish a one-time snapshot from sustained growth.

• Review payment method by gender, age group, category, and transaction value; use a cross-tabulation or statistical test before concluding that variables are related.

5. Limitations and Validation Notes

This report is based on a screenshot rather than the underlying dataset. Values are approximate where chart labels are small; currency, reporting period, and KPI definitions are not visible. The report avoids naming a definite top adult age group where the screenshot does not provide a reliable ranking. Validate exact numbers in Power BI before submitting or presenting to management.